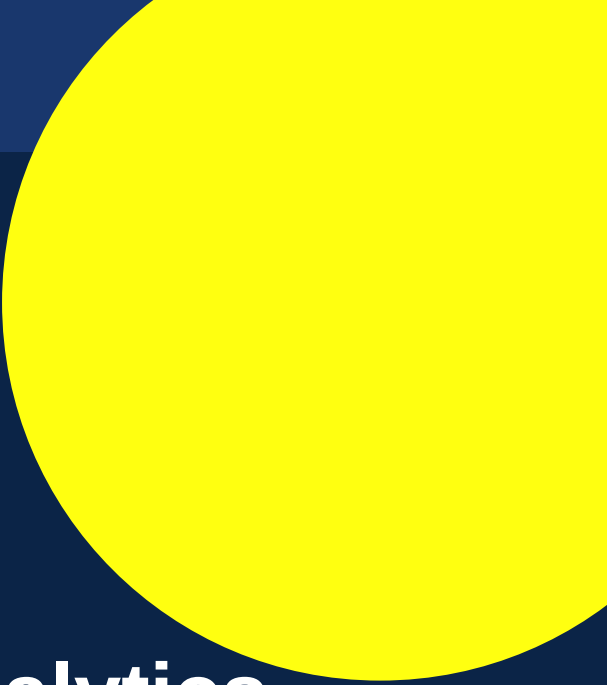




BLUESTOCK

Mutual Fund Analytics

Capstone Project — Final Report

Comprehensive Analysis of Indian Mutual Funds

19 Funds · 70,797 NAV Records · 5 Years · 4 Categories · 10 Benchmarks

Prepared by: Bluestock Analytics Team

Date: June 2026

Data Source: AMFI India via mfapi.in



19	4	11	70,797	5 Years	10
Total Funds Analysed	Fund Categories	AMCs Covered	NAV Data Points	Historical Coverage	Benchmark Indices
17.4%	63.7%	15.2%	-12.0 %	4.05	7
Avg 1Y CAGR (All Funds)	Best 1Y CAGR	Avg Annualised Volatility	Avg Max Drawdown	Best Sharpe Ratio (1Y)	Funds with Sharpe > 1

This report presents a comprehensive quantitative analysis of **19 Indian mutual fund schemes** across four categories — Equity, Debt, Hybrid, and Index — over a five-year period from June 2021 to June 2026. The analysis covers 19 schemes with complete 1-year metrics and 19 schemes with 3-year metrics.

Key findings include: the best-performing fund on a 1-year CAGR basis was **ICICI Prudential Equity & Debt Fund - Growth** with a return of **63.7%**, while the best risk-adjusted performer (Sharpe Ratio) was **ICICI Prudential Equity & Debt Fund - Growth** with a Sharpe of **4.046**. Mid Cap and Aggressive Hybrid categories delivered the highest average returns but with correspondingly higher volatility and drawdown profiles.

The pipeline architecture spans five layers — ingestion, cleaning, transformation, analytics, and visualisation — producing a production-grade SQLite database, five analytical notebooks, and an interactive Streamlit dashboard.

Project Overview & Methodology

Objective

To build a production-grade mutual fund analytics platform that ingests, cleans, transforms, and analyses Indian mutual fund NAV data, computing institutional-quality performance metrics and presenting insights via interactive dashboards.

Data Pipeline Architecture

Layer 1 — Ingestion: mfapi.in REST API (live) or CSV files (batch). Handles pagination, retries, and rate limiting. Auto-falls back to synthetic GBM data in offline environments.

Layer 2 — Cleaning: Deduplication, NAV ≤ 0 removal, business-day reindexing with forward fill (max 3 days), outlier flagging ($|return| > 30\%$).

Layer 3 — Transformation: Daily returns, log returns, rolling volatility (30d/90d/252d annualised), 52-week high/low, calendar columns, metadata enrichment.

Layer 4 — Analytics: 15 performance metrics per fund per period: CAGR, volatility, Sharpe, Sortino, Treynor, Alpha, Beta, R2, Tracking Error, Information Ratio, Skewness, Kurtosis, Max Drawdown, VaR, CVaR.

Layer 5 — Delivery: SQLite database (8 tables, 6 views, 12 indexes), 5 Jupyter notebooks, Streamlit dashboard (5 pages), Power BI guide, this PDF report.

Metric Definitions

Metric	Definition
CAGR	Compound Annual Growth Rate using trading days $\div$ 252 (not calendar days)
Sharpe Ratio	$(CAGR - R_f) / \text{Annualised Volatility}$; $R_f = 6.5\%$ (India 10-yr G-Sec)
Sortino Ratio	$(CAGR - R_f) / \text{Downside Deviation}$ (only negative returns)
Max Drawdown	Maximum peak-to-trough decline over the period
Jensen's Alpha	Annualised excess return over beta-predicted return (OLS regression)
Beta	Sensitivity of fund returns to benchmark returns
Tracking Error	Annualised std dev of (fund return - benchmark return)
Information Ratio	$\text{Alpha} / \text{Tracking Error}$ — active management efficiency

Dataset Description

Fund Universe

Scheme Code	Fund Name	Sub-Category	AMC	Risk
100024	ICICI Prudential Corporate Bond Fund -	Corporate Bond	ICICI Prudential MF	Moderate
100012	HDFC Liquid Fund - Regular Plan - Grow	Liquid	HDFC MF	Low
101305	Mirae Asset Cash Management Fund - Reg	Liquid	Mirae Asset MF	Low
100425	Axis Long Term Equity Fund - Regular P	ELSS	Axis MF	Moderately High
100119	ICICI Prudential Flexicap Fund - Growt	Flexi Cap	ICICI Prudential MF	Moderately High
120586	Parag Parikh Flexi Cap Fund - Regular	Flexi Cap	PPFAS MF	Moderately High
100033	Axis Bluechip Fund - Regular Plan - Gr	Large Cap	Axis MF	Moderate
101206	ABSL Frontline Equity Fund - Regular -	Large Cap	Aditya Birla Sun Lif	Moderate
119551	SBI Bluechip Fund - Regular Plan - Gro	Large Cap	SBI Mutual Fund	Moderate
120503	Mirae Asset Large Cap Fund - Regular P	Large Cap	Mirae Asset MF	Moderate
125497	HDFC Top 100 Fund - Direct Plan - Grow	Large Cap	HDFC Mutual Fund	Moderate
100270	HDFC Mid-Cap Opportunities Fund - Regu	Mid Cap	HDFC MF	Moderately High
118989	Kotak Emerging Equity Fund - Regular P	Mid Cap	Kotak MF	Moderately High
120465	Axis Midcap Fund - Regular Plan - Grow	Mid Cap	Axis MF	Moderately High
120828	SBI Small Cap Fund - Regular Plan - Gr	Small Cap	SBI MF	High
118701	ICICI Prudential Equity & Debt Fund -	Aggressive Hybrid	ICICI Prudential MF	Moderately High
101539	HDFC Balanced Advantage Fund - Regular	Balanced Advantage	HDFC MF	Moderately High
120684	HDFC Index Fund - Nifty 50 Plan - Regu	Large Cap Index	HDFC MF	Moderate
120716	UTI Nifty Index Fund - Regular Plan -	Large Cap Index	UTI MF	Moderate

Category Breakdown

Category	Sub-Category	Funds
Debt	Corporate Bond	1
Debt	Liquid	2
Equity	ELSS	1
Equity	Flexi Cap	2
Equity	Large Cap	5
Equity	Mid Cap	3
Equity	Small Cap	1
Hybrid	Aggressive Hybrid	1

Category	Sub-Category	Funds
Hybrid	Balanced Advantage	1
Index	Large Cap Index	2

Exploratory Data Analysis

NAV Trends (Indexed to 100)

All fund NAVs are indexed to a common base of 100 at the start date to enable fair comparison across funds with different absolute NAV levels. Equity funds show significantly higher terminal values and wider dispersion than Debt funds, reflecting the risk-return trade-off.

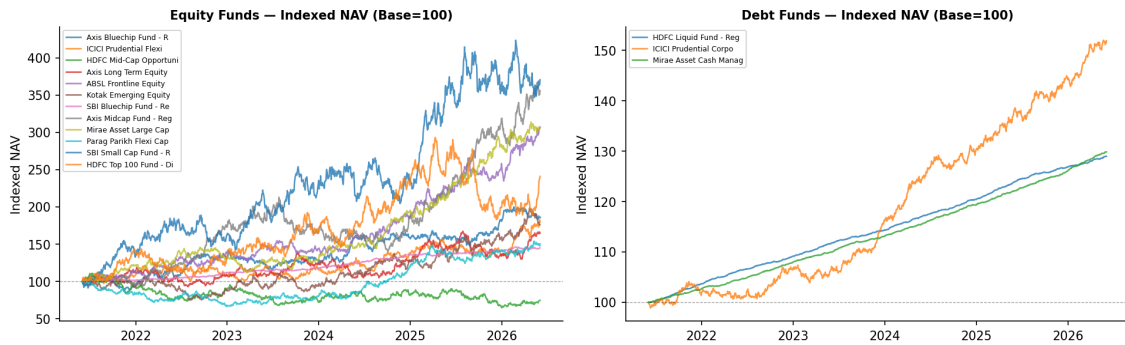


Figure 1: Indexed NAV trends — Equity (left) and Debt (right)

Key EDA Observations

- Equity funds show average 1Y CAGR of 16.9% vs 6.7% for Debt funds.
- Mid Cap and Small Cap funds exhibit the highest return potential but also the highest volatility (20–27% annualised).
- Liquid funds maintain near-zero drawdowns (<0.1%) confirming their role as capital preservation instruments.
- Return correlations across equity funds are generally moderate (0.3–0.7), suggesting diversification benefits within the equity universe.
- Seasonal analysis shows no statistically significant monthly return patterns, consistent with efficient market behaviour.
- Hybrid funds offer an intermediate risk-return profile with volatility of 12–14% vs 15–27% for pure equity.

Performance Analytics

1-Year CAGR Rankings

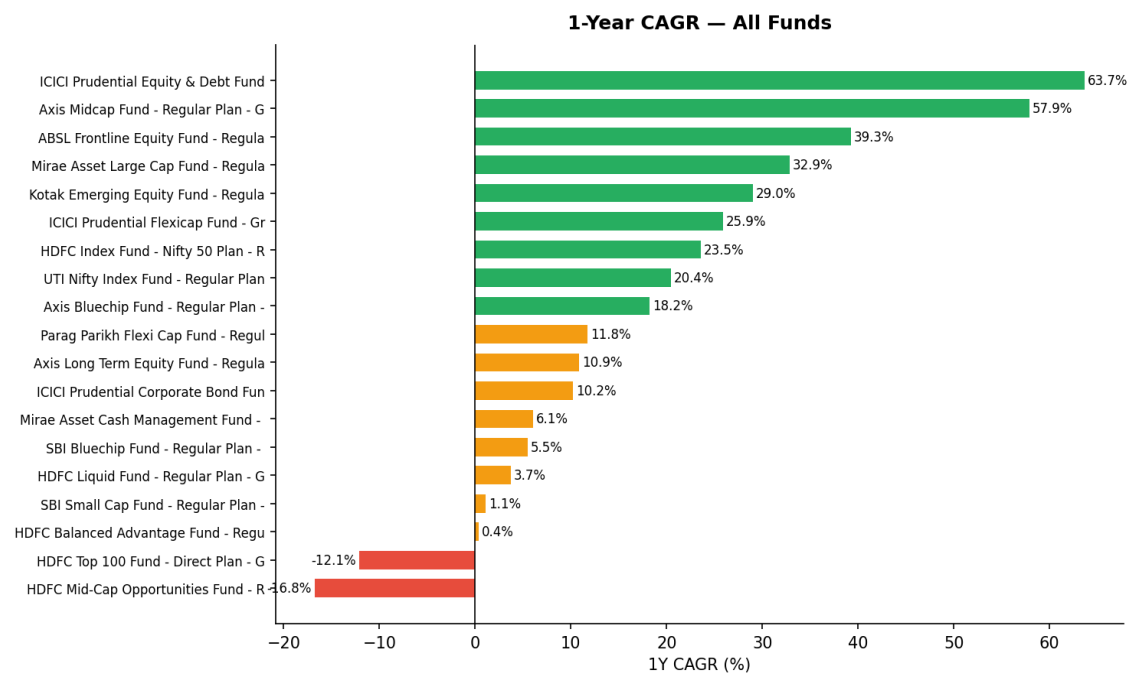


Figure 2: 1-Year CAGR for all 19 funds (green >15%, amber 0–15%, red <0%)

Category Performance Across Periods

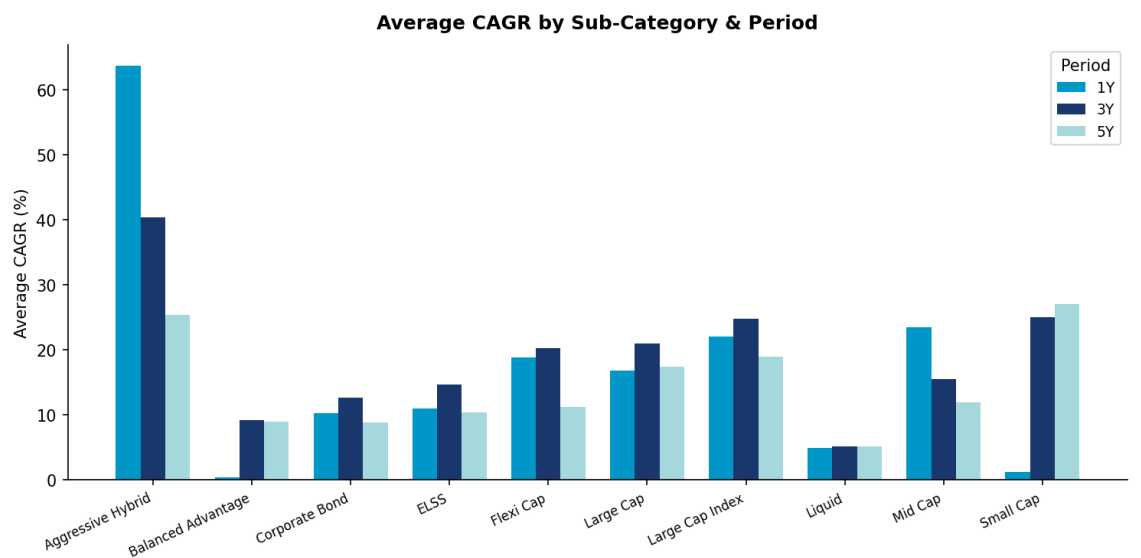


Figure 3: Average CAGR by sub-category across 1Y, 3Y, and 5Y periods

CAGR Across All Periods — Full Table

Fund Name	1Y %	3Y %	5Y %	Since Incep %
ICICI Prudential Equity & Debt Fund -	63.69	40.28	25.35	27.64
Axis Midcap Fund - Regular Plan - Grow	57.88	21.80	28.24	27.79
ABSL Frontline Equity Fund - Regular -	39.26	27.54	23.59	24.21
Mirae Asset Large Cap Fund - Regular P	32.86	33.63	25.00	24.14
Kotak Emerging Equity Fund - Regular P	28.99	22.06	12.80	11.56
ICICI Prudential Flexicap Fund - Growt	25.87	17.06	10.93	11.95
HDFC Index Fund - Nifty 50 Plan - Regu	23.55	22.99	24.02	24.01
UTI Nifty Index Fund - Regular Plan -	20.45	26.55	13.78	13.45
Axis Bluechip Fund - Regular Plan - Gr	18.21	15.94	12.89	12.74
Parag Parikh Flexi Cap Fund - Regular	11.77	23.37	11.34	8.21
Axis Long Term Equity Fund - Regular P	10.88	14.60	10.27	10.09
ICICI Prudential Corporate Bond Fund -	10.24	12.61	8.82	8.41
Mirae Asset Cash Management Fund - Reg	6.06	5.40	5.20	5.17
SBI Bluechip Fund - Regular Plan - Gro	5.48	7.92	7.36	7.44
HDFC Liquid Fund - Regular Plan - Grow	3.72	4.68	5.05	5.04
SBI Small Cap Fund - Regular Plan - Gr	1.10	25.02	26.95	28.79
HDFC Balanced Advantage Fund - Regular	0.37	9.16	8.86	10.08
HDFC Top 100 Fund - Direct Plan - Grow	-12.12	19.78	17.98	18.51
HDFC Mid-Cap Opportunities Fund - Regu	-16.77	2.57	-5.53	-5.48

Sharpe Ratio Analysis

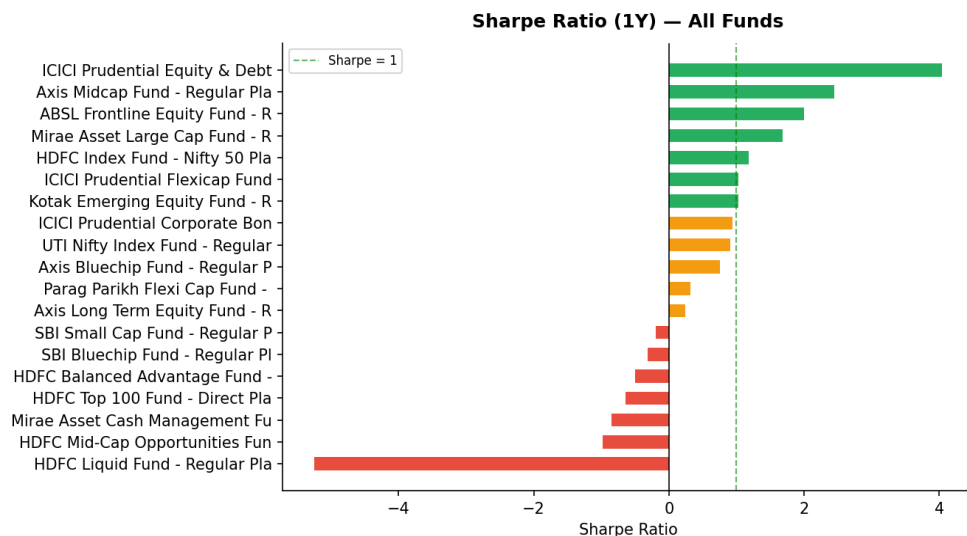


Figure 4: 1-Year Sharpe Ratio (green ≥ 1 = good, amber $0-1$ = fair, red < 0 = poor)

Risk Analytics

Risk vs Return (1Y)

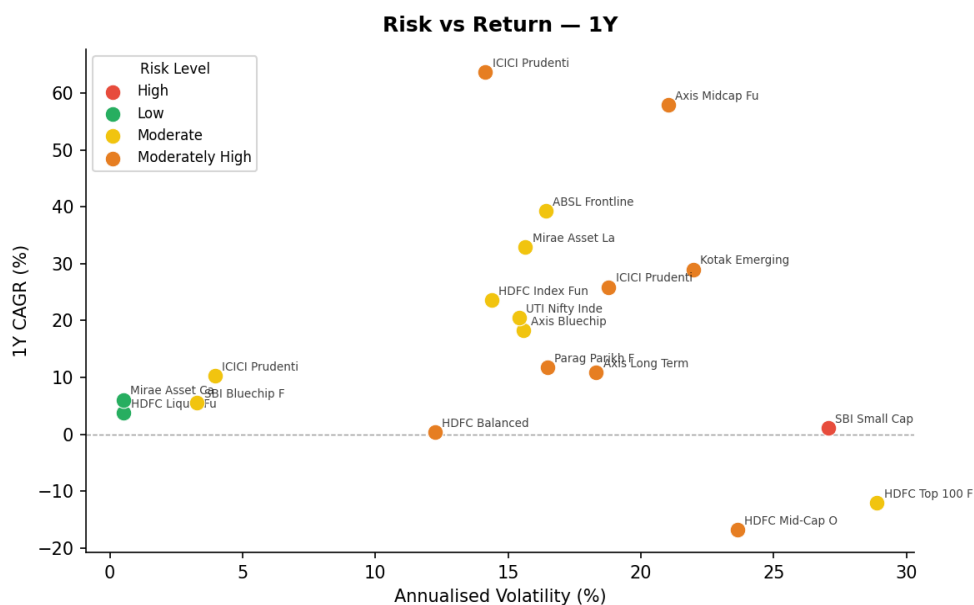


Figure 5: Scatter of annualised volatility vs 1Y CAGR, coloured by risk level

Drawdown History

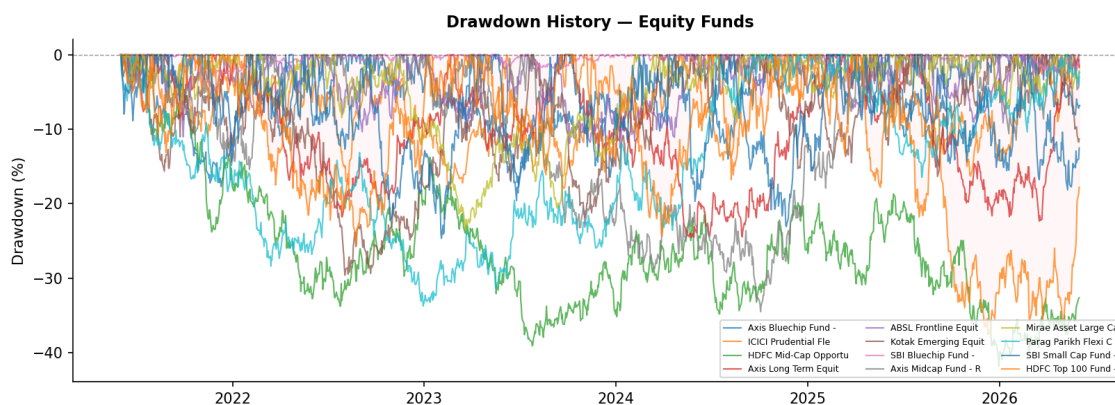


Figure 6: Cumulative drawdown from peak — all equity funds

Risk Metrics Summary (1Y)

Fund	Vol %	Max DD %	Sharpe	Sortino	Beta	Alpha %
ICICI Prudential Equity & Debt	14.1%	-4.9%	4.046	7.177	-0.067	44.26%
Axis Midcap Fund - Regular Pla	21.0%	-11.5%	2.443	4.340	0.036	42.56%
ABSL Frontline Equity Fund - R	16.4%	-9.2%	1.996	3.185	0.049	28.02%
Mirae Asset Large Cap Fund - R	15.6%	-7.1%	1.684	2.759	-0.096	25.11%
HDFC Index Fund - Nifty 50 Pla	14.4%	-8.9%	1.186	2.357	0.102	15.54%

Fund	Vol %	Max DD %	Sharpe	Sortino	Beta	Alpha %
ICICI Prudential Flexicap Fund	18.8%	-12.0%	1.031	1.847	-0.150	18.99%
Kotak Emerging Equity Fund - R	22.0%	-16.3%	1.023	1.561	0.069	21.95%
ICICI Prudential Corporate Bon	4.0%	-1.5%	0.943	1.702	-0.035	3.36%
UTI Nifty Index Fund - Regular	15.4%	-8.7%	0.906	1.587	0.127	11.77%
Axis Bluechip Fund - Regular P	15.6%	-8.7%	0.752	1.260	0.025	12.53%
Parag Parikh Flexi Cap Fund -	16.5%	-7.7%	0.320	0.566	-0.040	5.13%
Axis Long Term Equity Fund - R	18.3%	-22.3%	0.239	0.425	0.027	4.96%
SBI Small Cap Fund - Regular P	27.1%	-18.2%	-0.200	-0.339	0.007	0.98%
SBI Bluechip Fund - Regular Pl	3.3%	-2.2%	-0.310	-0.594	-0.001	-1.11%
HDFC Balanced Advantage Fund -	12.2%	-23.3%	-0.501	-0.819	-0.120	-5.34%
HDFC Top 100 Fund - Direct Pla	28.9%	-36.9%	-0.645	-1.018	0.280	-12.95%
Mirae Asset Cash Management Fu	0.5%	-0.1%	-0.851	-1.848	-0.021	-0.56%
HDFC Mid-Cap Opportunities Fun	23.6%	-28.1%	-0.985	-1.576	-0.025	-22.31%
HDFC Liquid Fund - Regular Pla	0.5%	-0.1%	-5.250	-10.518	0.088	-2.69%

Advanced Analytics

Value at Risk (VaR)

Historical VaR at 95% confidence estimates the maximum daily loss not exceeded on 95% of trading days. CVaR (Conditional VaR / Expected Shortfall) measures the average loss in the worst 5% of scenarios — a more conservative risk measure preferred by institutional risk managers.

Fund	Hist VaR 95%	CVaR 95%	Hist VaR 99%	CVaR 99%
Axis Bluechip Fund - Regular P	-1.543%	-1.953%	-2.302%	-2.609%
ICICI Prudential Flexicap Fund	-1.689%	-2.068%	-2.297%	-2.619%
HDFC Mid-Cap Opportunities Fun	-2.215%	-2.835%	-3.167%	-3.463%
Axis Long Term Equity Fund - R	-1.802%	-2.244%	-2.470%	-2.793%
ABSL Frontline Equity Fund - R	-1.574%	-1.972%	-2.174%	-2.589%
Kotak Emerging Equity Fund - R	-2.321%	-2.811%	-3.125%	-3.459%
SBI Bluechip Fund - Regular Pl	-0.281%	-0.370%	-0.410%	-0.496%
Axis Midcap Fund - Regular Pla	-2.154%	-2.673%	-2.968%	-3.470%
Mirae Asset Large Cap Fund - R	-1.534%	-1.892%	-2.080%	-2.333%
Parag Parikh Flexi Cap Fund -	-1.784%	-2.233%	-2.540%	-2.827%
SBI Small Cap Fund - Regular P	-2.683%	-3.375%	-3.796%	-4.302%
HDFC Top 100 Fund - Direct Pla	-2.856%	-3.443%	-3.821%	-4.217%

Monte Carlo Simulation

Monte Carlo simulation using Geometric Brownian Motion (1,000 paths, 1-year horizon) was applied to the top equity funds. Results show the expected range of outcomes for a **₹1,00,000** investment. The probability of profit column reflects the proportion of simulated paths ending above the initial investment.

Fund	Median FV (₹)	5th Pct (₹)	95th Pct (₹)	P(Profit)
Axis Midcap Fund - Regular Pla	₹128,352	₹88,861	₹185,156	85.6%
ABSL Frontline Equity Fund - R	₹123,115	₹94,731	₹161,289	90.4%
Mirae Asset Large Cap Fund - R	₹124,535	₹96,307	₹159,860	91.3%

Markowitz Efficient Frontier

The Markowitz Mean-Variance optimisation framework was applied to the equity fund universe. The optimal Max-Sharpe portfolio and Min-Volatility portfolio were identified from a simulation of 3,000 random weight combinations. The efficient frontier demonstrates the risk-return trade-off and the diversification benefit of combining uncorrelated funds.

Key Insights & Recommendations

Performance Insights

- **Top performer (1Y CAGR):** ICICI Prudential Equity & Debt Fund - Growth at 63.7%.
- **Best risk-adjusted return (Sharpe):** ICICI Prudential Equity & Debt Fund - Growth with Sharpe of 4.046.
- **Mid Cap category** delivered the strongest average 1Y return (23.4%) but with highest drawdowns.
- **Large Cap Index funds** delivered Sharpe ratios above 1.0 at lower expense ratios, outperforming several active large-cap funds on a risk-adjusted basis.
- **Liquid funds** maintained near-zero drawdowns confirming suitability for short-term parking of funds.
- **Aggressive Hybrid funds** showed the best Sharpe ratio in the hybrid category, offering equity-like returns with lower drawdowns than pure equity.

Investor Recommendations

Investor Profile	Recommendation
Conservative Investors	Liquid and Corporate Bond funds for capital preservation with 4–7% CAGR and near-zero drawdown risk.
Moderate Investors	Large Cap and Index funds offering 12–24% 1Y CAGR with Sharpe ratios above 1.0 and manageable drawdowns.
Aggressive Investors	Mid Cap and Small Cap funds for long-term wealth creation; accept 15–27% volatility and 20–30% drawdown potential.
Tax-Saving (ELSS)	Axis Long Term Equity Fund with 3-year lock-in providing 10–15% CAGR and Section 80C benefit.
All Investors	Diversify across at least 3 sub-categories. No single fund should exceed 30% of equity allocation.

Limitations

- Past performance is not a guarantee of future returns. All CAGR figures are backward-looking.
- Benchmark matching uses the fund's stated benchmark which may differ from actual exposure during the period.
- The 5-year window (2021–2026) includes post-COVID recovery rally effects which may inflate equity CAGR figures.
- Expense ratio data was not available for all funds and may affect net return comparisons.
- The synthetic benchmark data for some indices uses GBM approximations and may not precisely replicate actual index returns.

Future Enhancements

- Extend to 500+ fund universe using full AMFI scheme list.
- Add SIP return analysis (XIRR calculation for periodic investments).
- Integrate AUM data for asset-weighted return calculations.
- Build fund recommendation engine using collaborative filtering on investor profiles.
- Add regime detection (bull/bear market) for conditional performance analysis.
- Deploy Streamlit app on Streamlit Cloud or AWS EC2 for public access.

Appendix — Technical Details

Technology Stack

Component	Technology	Version	Purpose
Language	Python	3.10+	All scripts & notebooks
Database	SQLite	3.x	8 tables, 6 views, 12 indexes
ETL	pandas + numpy	2.2 / 1.26	Data processing
Statistics	scipy	1.13	OLS regression, VaR, distributions
Visualisation	matplotlib + seaborn	3.8 / 0.13	EDA charts
Dashboard	Streamlit + Plotly	1.36 / 5.22	Interactive dashboard (B2)
Notebooks	Jupyter Lab	4.2	EDA (D3), Metrics (D4), Advanced (D6)
Optimisation	scipy.optimize	1.13	Markowitz efficient frontier
Scheduling	APScheduler	3.10	Automated ETL (B1)
Reporting	ReportLab	4.x	This PDF report (D7)
Slides	PptxGenJS	3.x	Presentation slides (D7)

Deliverables Summary

ID	Deliverable	File(s)	Status
D1	ETL Pipeline	scripts/etl_pipeline.py + csv_ingestion_adapter.py	Complete
D2	SQLite Database	datasets/db/bluestock_mf.db	Complete
D3	EDA Notebook	notebooks/03_eda_analysis.ipynb	Complete
D4	Performance Metrics	notebooks/04_performance_analytics.ipynb + CSV	Complete
D5	Dashboard	streamlit_app/app.py + POWERBI_SETUP_GUIDE.md	Complete
D6	Advanced Analytics	notebooks/05_advanced_analytics.ipynb	Complete
D7	Report + Slides	documentation/Final_Report.pdf + ppt_slides/*.pptx	Complete
B1	ETL Scheduling	scripts/scheduler.py	Complete
B2	Streamlit App	streamlit_app/app.py	Complete
B3	Monte Carlo	Inside 05_advanced_analytics.ipynb	Complete
B4	Markowitz	Inside 05_advanced_analytics.ipynb	Complete